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Course: Cloud Computing

1. Unified Big data Architecture for multinational Retail Organisation:

A multinational retail organisation may collect customer data from website, mobile applications, physical stores, loyalty program, social media & payment system. If these resources are isolated, the organisation cannot build a complete customer profile or provide consistent personalised experience.

Benefits:

- Complete 360 degree view of customer.
- Real time analysis of customer behaviour.
- Personalised product recommendation.
- Better marketing campaigns.

2) Distributed computing for scientific Research data Concept:

Scientific research generates huge amount of data in areas such as climate studies, space research, medical research and simulation.

1) Multimedia data Management in Big Data

Challenges :

Multimedia data makes the storage, backup & handling.

A single computer may take very long time to process the data.

Distributed Computing Models :

- * Hadoop Map Reduce
- * Apache Spark
- * High performance computing (HPC)
- * Cloud-based distributed systems.

2).

Fraud Detection Framework for Digital Payments.

Framework :

Transaction Data $\rightarrow$ Streaming platform $\rightarrow$ Realtime processing $\rightarrow$ ML Model $\rightarrow$ Fraud Detection $\rightarrow$ Alert.

Working :

- * Payment transaction are continuously collected
- * Streaming tools process transaction in real time
- * Features such as location, transaction amount, device, and time are also analysed.
- * A machine learning model predicts whether the transaction is genuine or fraudulent.
- * Suspicious transaction are blocked or sent for verification.

* Reduced Trust in analytics

4) Multimedia data Management in Big Data: Challenges :

Multimedia data such as videos, images and audio files creates several problems.

- * Very large file sizes
- * High storage requirements :
- * Different formats.
- * Slow processing
- * Difficult searching & indexing.
- * High bandwidth requirements.

Benefits :

- * Scalable storage
- * Faster processing
- * Easy access
- * Improves multimedia management.

5) Disaster Recovery & Business continuity strategy Objective :

The main goal is to keep the big data platform operational even during server failures, cyberattacks, natural disasters or data center outages.

Architecture

Primary cloud $\rightarrow$ Data Replication $\rightarrow$ secondary
Region $\rightarrow$ Automatic failover.

Benefits:

- * Reduced downtime
- * Better data protection.
- * Continuous business operation
- * Faster recovery after disaster.

6). Data Quality and Data Governance -
Common Data Quality Problems

* Inconsistency ..

The same data may appear in different formats.

Eg: India, IND and IN

* Duplication ..

The same customer recorded stored multiple times.

* Missing Values ..

Important information such as phone numbers or addresses may be absent.

Impact ..

$\rightarrow$ Impact reports

$\rightarrow$ Wrong predictions.

Dashboard & Analytics results:

Architecture:

- * Real decisions on Journeys
- * Reduced Latency in analytics

7) Scalable Recommendation systems for online shopping.

Architecture:

User Activity $\rightarrow$ Data Collection $\rightarrow$ Big data storage $\rightarrow$ Distributed processing $\rightarrow$ ML Model $\rightarrow$ Personalized recommendations

Working:

The system analyses behaviour of millions of users if it identifies similar users and recommends content based on viewing patterns & preferences.

Benefits:

- * More accurate recommendations.
- * Handles millions of users.
- * Provides personalized experience.

8) Role of cloud computing in Big Data:

Cloud computing provides computing resources such as storage, servers through the internet.

Limitations :

- * Data Security Concerns
- * Vendor Dependency
- * Unencrypted data
- * Vendor lock-in
- * Limited control over infra structure.

Evaluation :

Cloud computing plays an important role in big data because it allows organizations to process massive datasets without investing heavily in physical infrastructure.

Real time analytics for Transportation and Logistics

Framework :

GPS / Sensor → Data streaming → Real-time processing → Analytics Engine → Dashboard Decision Making

Technologies :

- Apache, Kafka → Collect Streaming data
- Apache spark streaming → process data
- Cloud storage → Store historical data
- Machine Learning → Predict traffic & etc
- Dashboard → Display results

→ Dashboard & Display results.
Applications:

- Route Optimized.
- Fuel Efficiency.
- Operational Decision.

10) Ethical & Legal Challenges in Big Data,
Ethical Challenges:

- * Privacy
- * Data Mining
- * Algorithmic Bias.
- * Lack of Transparency.

Legal Challenges:

- * Data protection laws.
- * Privacy regulation.
- * User consent requirements.
- * Data security standards.